

# Purplle.com: Boosting workflows to launch hundreds of beauty products a month with Google Cloud

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## About Purplle

Purplle uses Google Cloud to streamline infrastructure management and data analytics and cloud technology to improve workflow and enhance customer experience.

Purplle is an online space for beauty and wellness needs that aims to take its customers' shopping experience to a completely different level. The company averages four million monthly users in India, on its app and site, including tier two and three markets. By adopting analytics and cloud technology, Purplle has streamlined infrastructure management and sees a healthy growth in revenue.

Industries: Retail & Consumer Goods

## Google Cloud results

- Achieves 100% year-on-year growth in terms of revenue
- 50% cost savings on server and 75% reduction in production cost
- Enables the launch of 150 products a month
- Collect up to 40 million event data points a day

## Support reducti produc

Location: India

(<https://cloud.google.com/dataflow/>), [Cloud](#)

[Google Cloud](#) (<https://cloud.google.com/>)

(<https://cloud.google.com/dlp/>), [Cloud](#)

[App Engine](#) (<https://cloud.google.com/appengine/>)

(<https://cloud.google.com/datastore/>), [Cloud](#)

[BigQuery](#) (<https://cloud.google.com/bigquery/>)

[Cloud Bigtable](#) (<https://cloud.google.com/bigtable/>)

(<https://cloud.google.com/iam/>), [Cloud](#)

[Dataflow](#) (<https://cloud.google.com/dataflow/>)

[Cloud SQL](#) (<https://cloud.google.com/sql/>)

[Cloud Data Loss Prevention](#)

(<https://cloud.google.com/dlp/>)

[Cloud Datastore](#)

(<https://cloud.google.com/datastore/>)

[Cloud Functions](#)

(<https://cloud.google.com/functions/>)

[Cloud IAM](#) (<https://cloud.google.com/iam/>)

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[Cloud Pub/Sub](#) (<https://cloud.google.com/pubsub/>)

[Cloud SQL](#) (<https://cloud.google.com/sql/>)

[Cloud Storage](#) (<https://cloud.google.com/storage/>)

[Firebase](#) (<https://firebase.google.com/>)

Founded in 2012, [Purple.com](https://www.purple.com/)

(<https://www.purple.com/>) is one of the largest online

beauty retailers in India, serving millions of

customers every year in a market that's as complex as it is promising. Specific products drive the search for goods like mobile phones; but in beauty and hygiene, the country's third most searched vertical (<https://yourstory.com/2016/02/beauty-hygiene-market>), search takes the form of open-ended questions. This discovery-oriented market leads to unique challenges in bringing up the right products for customers' specific needs and accommodating highly personal customer preferences.

The huge amount of data and information makes high-inventory online retailers like Purplle difficult for users to navigate. As a result, the company suffered from low conversion rates as the platform appeared to lack certain products and differentiators.

An added challenge for the online retailer was how to achieve hyper-personalization, which leverages artificial intelligence (AI) and real-time data to deliver relevant product and service information to each user.

"Beauty is a complicated category because there are many options available for the same product. You can't touch or feel a product that you're buying online, which leaves room for uncertainty," says Suyash Katyayani, Chief Technology Officer and co-founder of Purplle.

In late 2017, Purplle decided to personalize the browsing experience for its customers. "We didn't want to base the experience just on online user behavior or the availability of products. We wanted

to tackle personalization based on user persona, affinity, and taste,” he says.

As hyper-personalization requires real-time data mining and content serving, Purplle, which relied on several cloud services to achieve low-latency data mining, saw a need to migrate to a bigger provider.

## **A smooth sailing migration process to Google Cloud**

In November 2017, Purplle chose [Google Cloud](https://cloud.google.com/) (<https://cloud.google.com/>) for its digital transformation journey. Suyash adopted the 80/20 rule for the cloud migration, where migration teams spend 80% of their effort planning so that the execution requires only 20% of the effort.

The migration from Purplle’s previous, smaller cloud provider to Google Cloud took just three days. It ran a parallel server on Google Cloud that was up to date with the latest data and code changes. As a result, users experienced only five to ten minutes of actual downtime while the connection was switched to the new server.

“Previously, all our servers were homogenous, and we faced operational overheads in terms of server management. We felt that too much time was spent managing infrastructure instead of our core business, which prompted us to look at cloud infrastructure,” Suyash says.

Before switching to Google Cloud, Suyash engaged an infra-management company to manage and

monitor the company servers. “We had to hire people who could set up and run the infrastructure, rather than build applications on top of it,” he says. After migration to Google Cloud, the overheads were eliminated.

“One of the biggest improvements I’ve seen in my product and technical team after migrating to Google Cloud is the turnaround time. Nothing is impossible, because you have state-of-the-art resources at your fingertips.”

—Suyash Katyayani, Chief  
Technology Officer and  
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Purplle also received strong support from Google as it participated in the Google Sand Hill Program, which offers product support and partnership opportunities for high-growth companies.

“The program gave us additional access to resources and advice on how to plan our migration. We created a system on Google Cloud and the deployment was done on Google Cloud servers. We migrated the application, database, and raw data capturing according to a straightforward process. We also did a data sync and replicating master and slave, which took us a day to complete. It was easy because of the proper planning beforehand.”

Since its migration to Google Cloud, Purplle has been enjoying the benefits of serverless technology. “One of the biggest improvements I’ve seen in our product and technical team after switching to Google Cloud is the turnaround time. Nothing is impossible, because you have state-of-the-art resources at your fingertips.”

With Google Cloud, Purplle’s team can experiment with application development without resource constraints. The team recently used [Firebase](https://firebase.google.com/) (<https://firebase.google.com/>) to enable a chat function in two weeks. Chat is now another channel for valuable customer insights at Purplle and its Chatbot Beauty Assistant generates 75,000 customer interactions a month.

Purple - Mumbai Summit Customer...



## **Achieving hyper-personalization with Cloud Bigtable**

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Managing infrastructure is one of the biggest challenges that Purplle faces in the journey of transformation. Hyper-personalization is one good way to connect with customers, and the company has identified this as an important opportunity.

To offer customized recommendations, Purplle requires a database that serves the same latency to different users. “We use Cloud Bigtable (<https://cloud.google.com/bigtable/>) to serve personalized recommendations to the user within milliseconds. As Cloud Bigtable helps to manage multiple downstreams of data, creating a data pipeline is simple for us,” he adds.

On top of this, Purplle leverages the managed services provided by Google Cloud. Without having to worry about IT overheads, the team can now



focus on other initiatives such as product development.

For instance, Purplle has pioneered the successful creation of 10 digitally native beauty brands including Good Vibes, Stay Quirky, and NY Bae. Purplle uses the platform data to analyze where the gaps are and create the supply chain pipeline. “Big brands launch four to five products a year, but we’re able to launch 150 new products a month.”

## **BigQuery: Data warehouse for big event data collection**

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For Purplle, BigQuery.

(<https://cloud.google.com/bigquery/>) acts as a data warehouse for behavioral events as well as for all the batch processing and customer 360-degree views, which Purplle uses for customer segmentation and product analytics.

Within Google Cloud, Purplle uses BigQuery to drive product analytics, user segmentation, and reporting for the business teams, which helps the company drive actionable insights and share the right metrics to the right business owners.

“We were unable to do many product launches due to a lack in supply differentiation. When we switched to Google Cloud in 2017, we saw a 100% year-on-year growth in terms of revenue,” Suyash says.

Using the analytics from BigQuery, Purplle has achieved great success in the number of product launches, from five to 200 new product launches a month. Overall conversion rate has improved by 45% and doubled for users with beauty personas due to the personalized experience the company is now able to offer. Retention on the app reached 52% within 30 days, and Purplle has managed to retain customers by providing them with a differentiated shopping experience.

Purplle also uses Cloud Pub/Sub

(<https://cloud.google.com/pubsub/>) and Cloud Dataflow (<https://cloud.google.com/dataflow/>) to build its data pipeline, which helps the company to collect up to 40 million event data points a day, or around 1.2 to 1.5 billion event data points on a monthly basis.

“Storing the data on Google Cloud is cost effective and has reduced production costs by 75%, as the job can be done by one person instead of a four-man team. On servers, we achieved a 50% cost saving by switching to Google Cloud.”

## **Ensuring security thanks to Cloud IAM**

Purple is also cautious about protecting customer data, handling it securely and ensuring that staff follow strict internal protocols. “Any identifiable user information in our user events that we collect is not stored on BigQuery. Any access to the personally identifiable information in the application database is encapsulated using application logic and Cloud IAM (<https://cloud.google.com/iam/>) (Identity Access Management),” Suyash says. “We only share user data which is anonymized. We respect user privacy and follow the best practices on personal identifiable data.”

Suyash notes that built-in protection from Google Cloud helps to prevent malicious bot or spyware from attacking the servers. Cloud IAM gives Purple full control and visibility to manage cloud resources centrally and provides a firewall to block server access for unwanted ports and IPs.

## **Experimenting with augmented reality using Vision AI**

Purple is experimenting with API technology and hopes to enhance the online buying experience using augmented reality. For instance, the company

is using Vision AI (<https://cloud.google.com/vision/>) to create a “skincare selfie” analysis feature that visually educates users about their skin concerns. “We created a lipstick match feature where the user clicks on her selfie and our system detects the facial coordinates using Vision AI, then recommends a matching lipstick color,” Suyash explains.

More augmented reality experiments are in the pipeline. “We’re expanding the live video ‘try-on’ feature to 10 more categories, including foundation, eyeshadow, and blush,” Suyash says.

Moving forward, PurpIle is keen to explore AutoML Tables (<https://cloud.google.com/automl-tables/>), Container Registry (<https://cloud.google.com/container-registry/>) and Google Kubernetes Engine (<https://cloud.google.com/kubernetes-engine/>) to bring the business to a new level.

“Kubernetes would help us organize our complex applications and enable us to be more efficient and agile in our deployment process,” shares Suyash. “We want to create a CI/CD pipeline to help us automate steps in our software delivery process, such as initiating code builds, running automated tests, and deploying to a staging or production environment.

He adds: “We’re hoping to make state-of-the-art machine learning models available to our business teams for them to run predictions such as demand forecasting and customer churn. AutoML Tables is an exciting product for us.”



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